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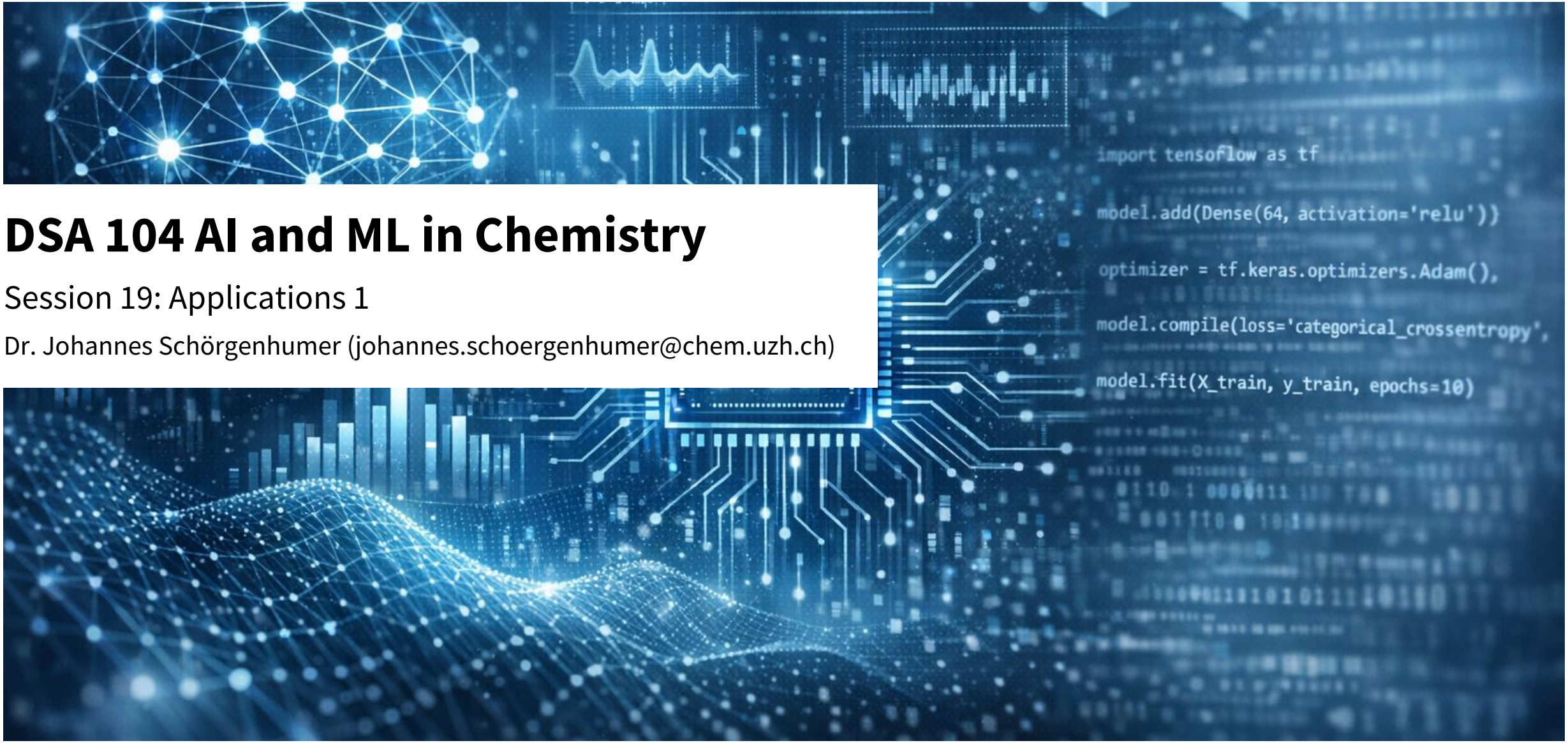
Department of Chemistry



DSA 104 AI and ML in Chemistry

Session 19: Applications 1

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Points for today

Discussion assignment 7

Applications of ML in chemistry:

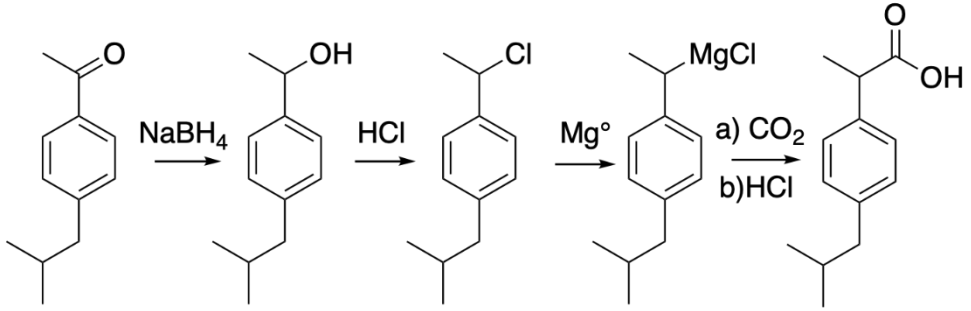
— Reaction prediction

— Retrosynthesis

Synthesis and retrosynthesis: A view on reactions

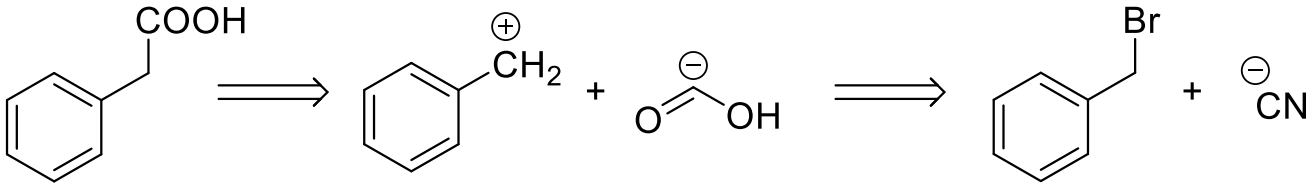
Synthesis

— Finding an efficient sequence of reactions to produce a target molecule



Retrosynthesis

— Breaking down target molecules into smaller parts by feasible pathways until you reach available materials (“disconnection approach”)



Different directions and different goals, but…

— **Similar tasks**

— Similar underlying **mappings** (reactants to products)

—What would be forward-oriented tasks / questions?
—What would be some retrosynthesis tasks / questions?

Discuss how you would translate them into ML.

Ibuprofen synthesis scheme by Leafsfan3 - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=141651824

Why ML matters in chemical synthesis / retrosynthesis

— Chemical space: estimated 10^{60} to 10^{100} molecules

— Synthesis = navigating chemical space combinatorially

Manual retrosynthesis / synthesis planning:

— Relies on expert heuristics

— Time-consuming and biased

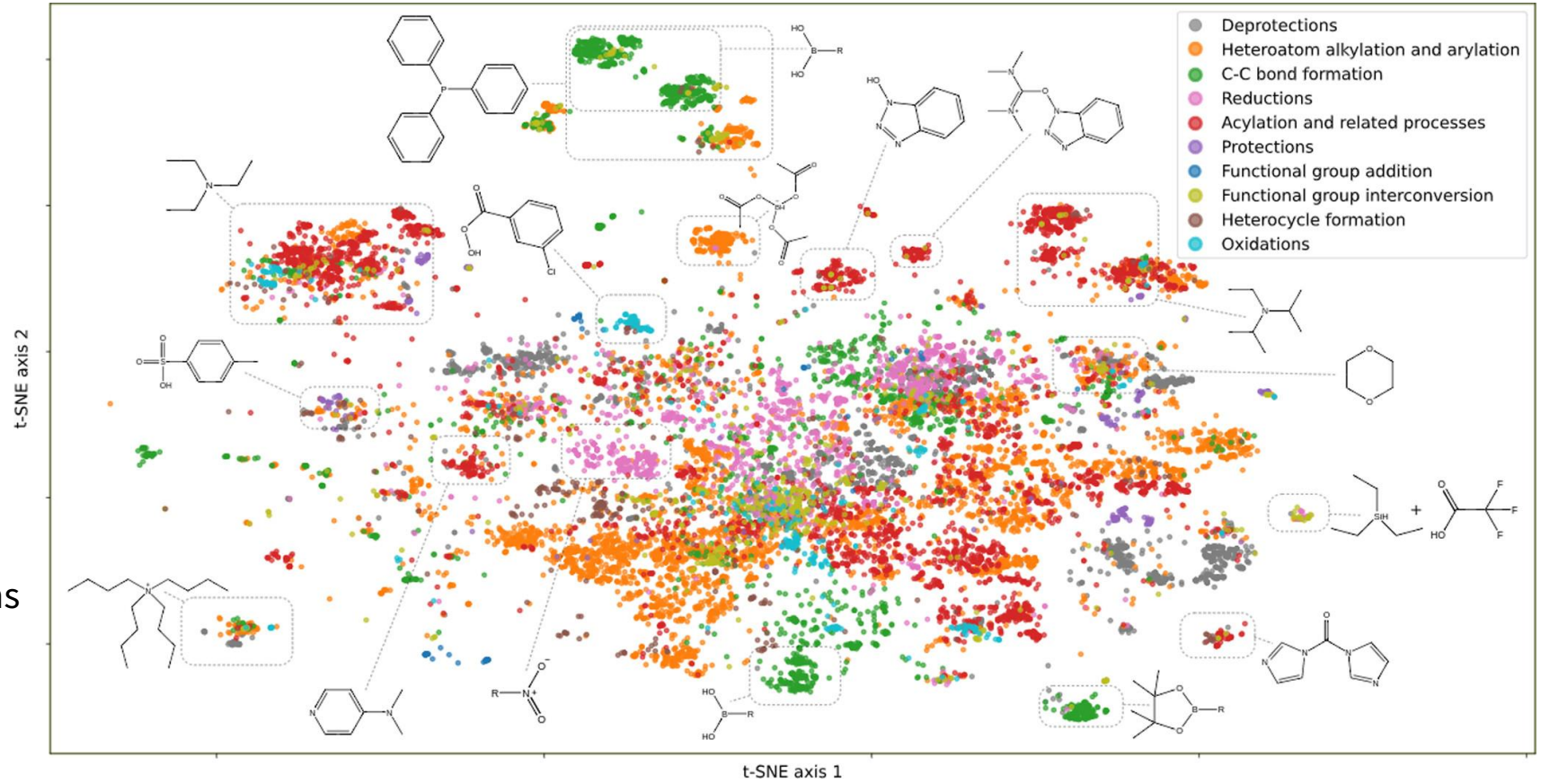
ML advantages:

— Learns patterns from millions of reactions

— Scalable

— Consistent

— Fast



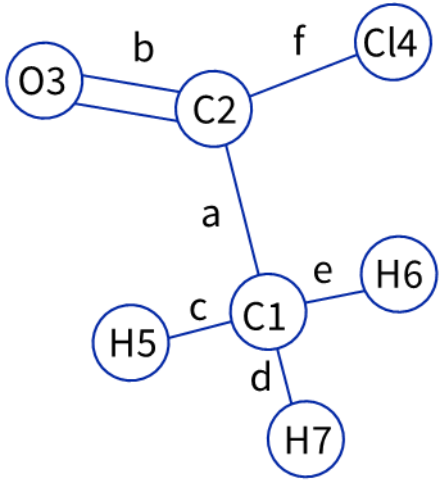
Scheme for reactions space: https://pubs.acs.org/doi/10.1021/acsomega.1c04778

Representations of chemical reactions

Molecules:

— Graphs

— Strings



CC(=O)Cl

Reactions:

— Graph transformations

— String translations

— Mappings:

  — Reactants, reagents, solvents, conditions → products

  — Often simplified: reactants → products

Conditions:

— Numerical features

Compare:

Unmapped reaction:
CCO >> CC=O

Mapped:
[C:1][C:2][O:3] >> [C:1][C:2]=[O:3]

Predicting a reaction outcome

One task of reaction prediction is to predict the reaction outcome.

Different approaches are possible:

Formulation	Input	Output
Classification	Reaction fingerprint	Product class
Sequence-to-sequence	SMILES	SMILES
Graph-to-graph	Molecular graphs	Molecular graph

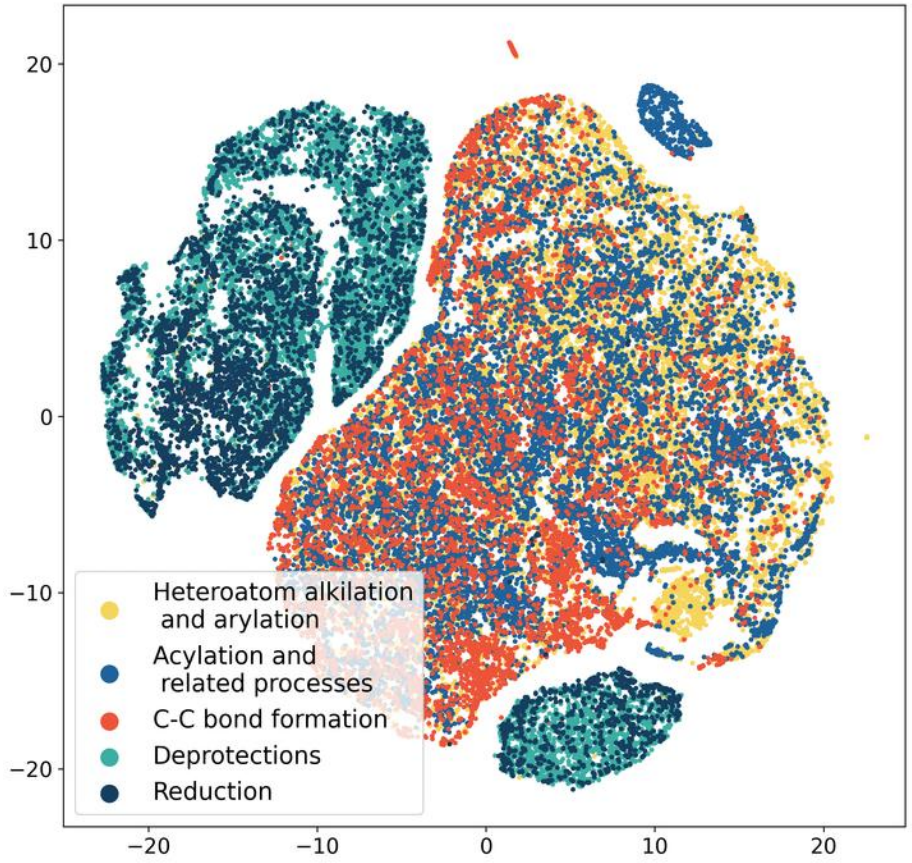
Have a look at the multiclass classification approach in the *ReactionPrediction.ipynb* notebook.

What is the main limitation?

Reaction learning: Datasets

Note: USPTO datasets do not include full reaction conditions, uncertainty unavoidable.

- USPTO-MIT
  - ~480k samples, clean and curated
  - Classic choice for reaction prediction
  - For classification, seq-to-seq and ranking
- USPTO-50k: Lower diversity, quick ML experiments
- Full USPTO: Full-scale (~2M reactions), realistic challenges (minimal curation, noise)
- USPTO-LLM:
  - ~250k reactions as graphs
  - Includes temperature, solvent, time
- ORD: Experimental realism:
  - Rich metadata (conditions, scale, yield)
  - Focus on structured HTE datasets
  - Smaller, but experimentally very powerful



TSNE representation of reaction space in USPTO-50k: https://chemrxiv.org/doi/full/10.26434/chemrxiv.12594785.v2

Synopsis: Synthesis planning

Different tasks, different directions:

Task	ML Role
Outcome prediction	Forward model
Retrosynthesis	Inverse problem
Planning	Search using both

Good synthesis planning needs both directions!

Next: look at **retrosynthesis**!

ML in retrosynthesis: Template-based approaches

Single step retrosynthesis:

Given a product molecule, predict the best disconnection

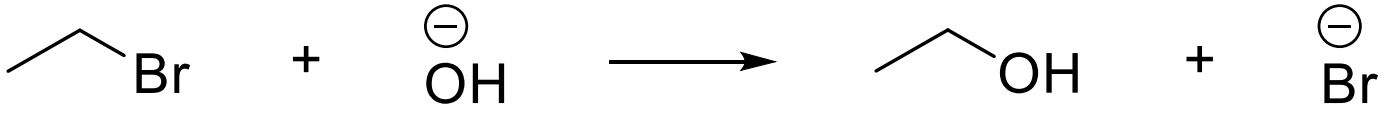
— Input: product graph

— Output: reaction **template** or reactant set

— ML framing: Multiclass classification (~10k–100k templates)

Pipeline for template-based approach:

Product
↓
Reaction class (how to cut)
↓
Reaction template (where to cut)
↓
Reactants



SMILES: CCBr.[O-] >> CCO.[Br-]

Template (SMARTS/SMIRKS): [CX4]-Br >> [CX4]-OH

ML in retrosynthesis: Template-based approaches

Learn **which reaction templates apply**

Pros:

— Chemically interpretable

— Valid reactions

Cons:

— Template extraction bias

— Limited creativity

Typical models:

— Feedforward nets (**FFNs**)

— Graph neural networks (**GNNs**)

ML in retrosynthesis: Template-free approaches

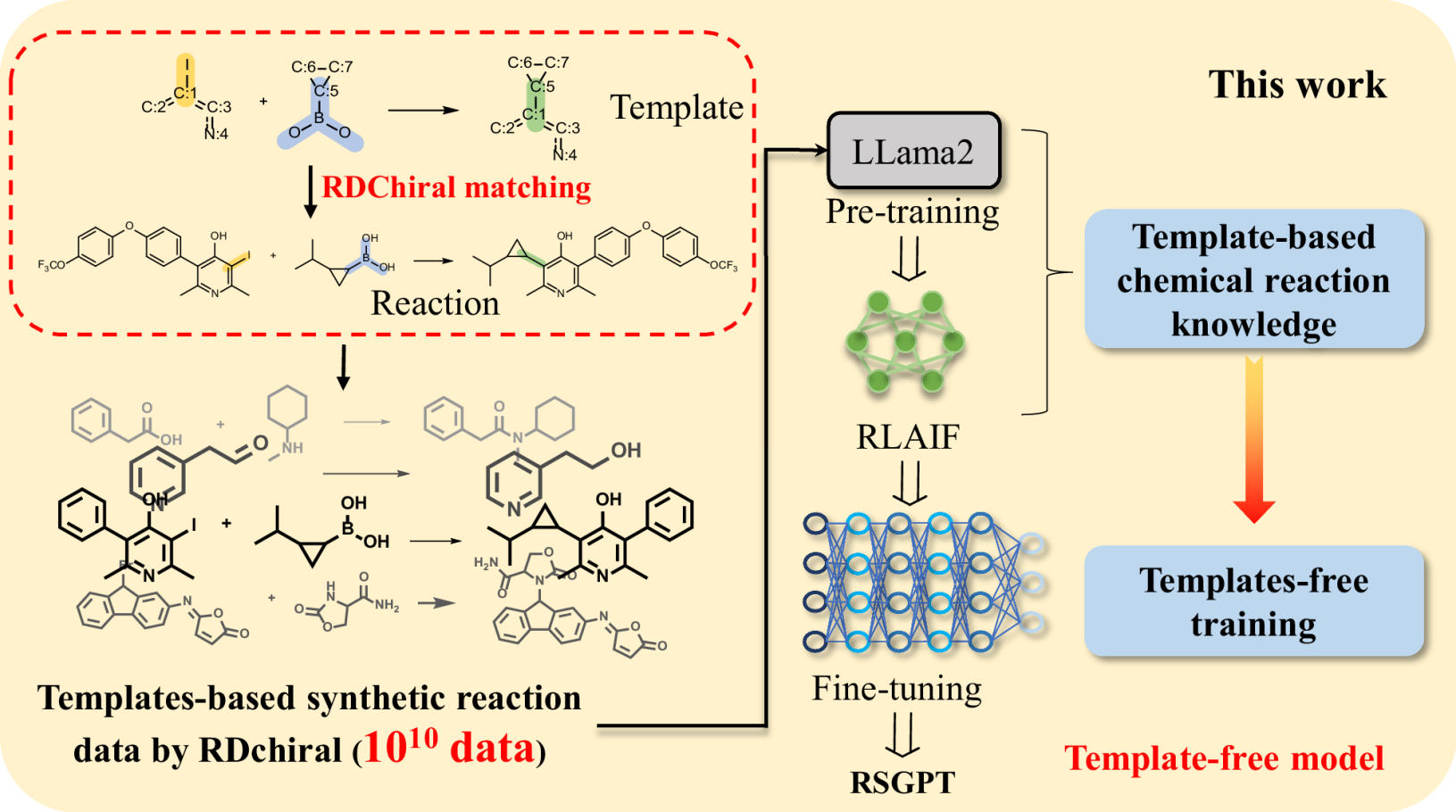
Generate reactants autoregressively

— Similar to machine translation:

  — Product SMILES → Reactant SMILES

— Main limitation according to literature: **Limited data**.

— E.g. transformer trained on billions of datapoints: ~63% top-1 accuracy



Deng, Y., Zhao, X., Sun, H. *et al.* RSGPT: a generative transformer model for retrosynthesis planning pre-trained on ten billion datapoints. *Nat Commun* **16**, 7012 (2025). https://doi.org/10.1038/s41467-025-62308-6

ML in retrosynthesis: Template-free approaches

Treat reactions as **translations**

Pros:

— More flexible

— Learns chemistry implicitly

Cons:

— Less guaranteed validity

Common models:

— **Transformers**!

ML in retrosynthesis: Multi-step approaches

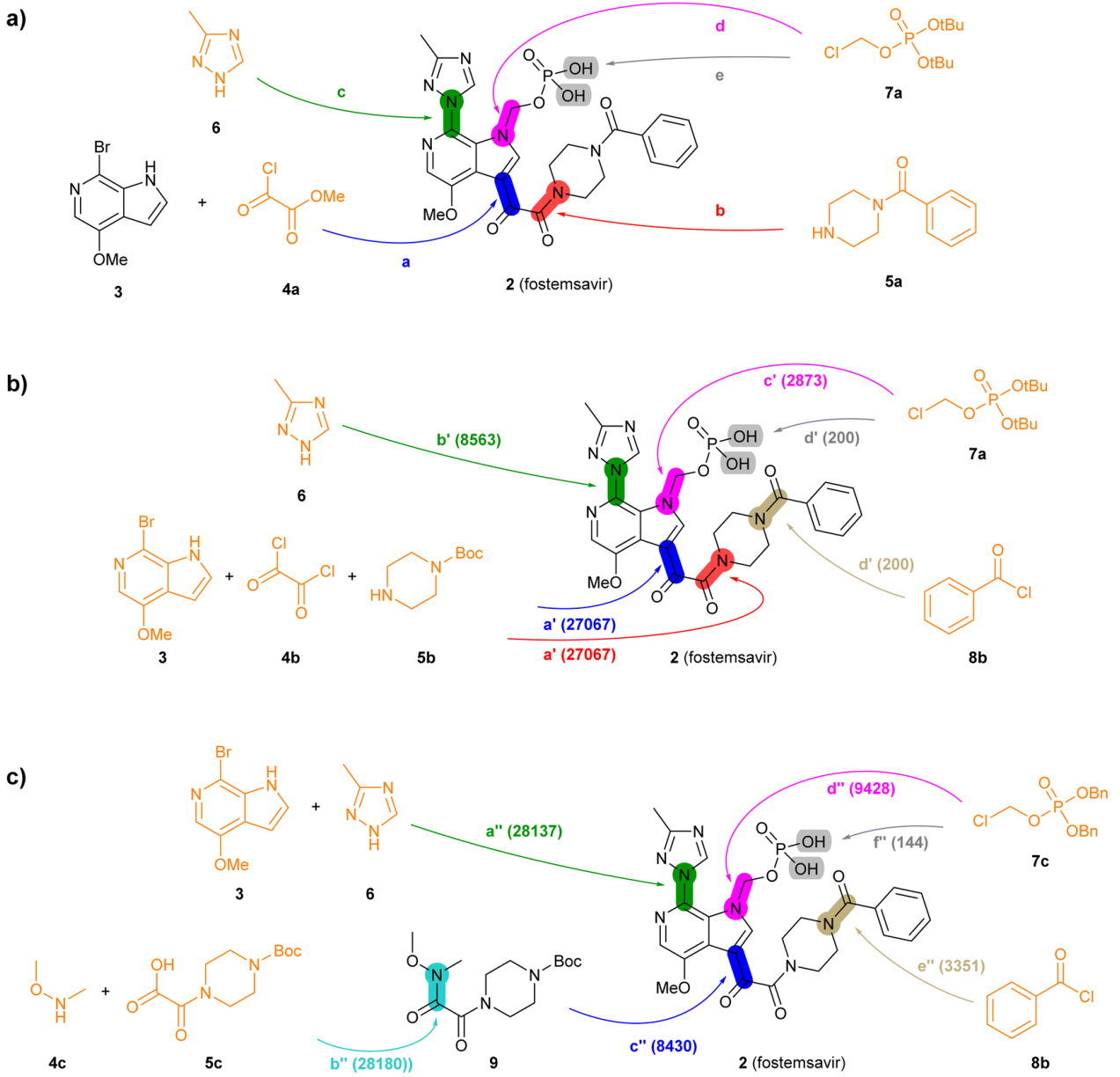
Multi-step synthesis planning (search):

Retrosynthesis as a tree search

— Actions: apply reaction rules

— Goal: reach purchasable compounds

— Often: Reinforcement learning with Reward systems, e.g. Route penalty score.



D. Kreutter, J.-L. Reymond, *Chem. Sci.*, **2023**, *14*, 9959-9969. DOI: 10.1039/D3SC01604H

ML in retrosynthesis: Multi-step approaches

Multi-step planning algorithms

Monte Carlo Tree Search (MCTS)

— Node = molecule

— Edge = reaction

— Policy network: which reaction to try

— Value network: how promising a molecule is

Reinforcement learning!

— State: current set of molecules

— Action: choose reaction

— Reward, e.g.: reach purchasable starting materials quickly

Wins and Fails for ML in retrosynthesis

Strengths:

— Learns non-obvious disconnections

— Scales to massive databases

— Reduces bias toward “textbook” reactions

Limitations:

— Garbage-in-garbage-out training data

— Poor modelling of:

  — Yields

  — reaction conditions

  — Stereochemistry

— Model confidence ≠ chemical feasibility

Retrosynthesis exercise

Look at the *Retrosynthesis.ipynb* notebook.

Compare the code with the reaction prediction.

Run the code and investigate the prediction of the reaction class (first part)

Then in the second part, see what suggestions you get for actual reactants.

Try with some molecules of your own and discuss the outcome / limitations.

Example: Chematica

Rule-based approach:

Grzybowski, Bishop et al., *Angew. Chem. Int. Ed.*, **2012**, *51*, 7928-7932. https://doi.org/10.1002/anie.201202209

— Chemistry software developed in 2012

— Algorithms to shorten pathways and for parallel optimisation of syntheses

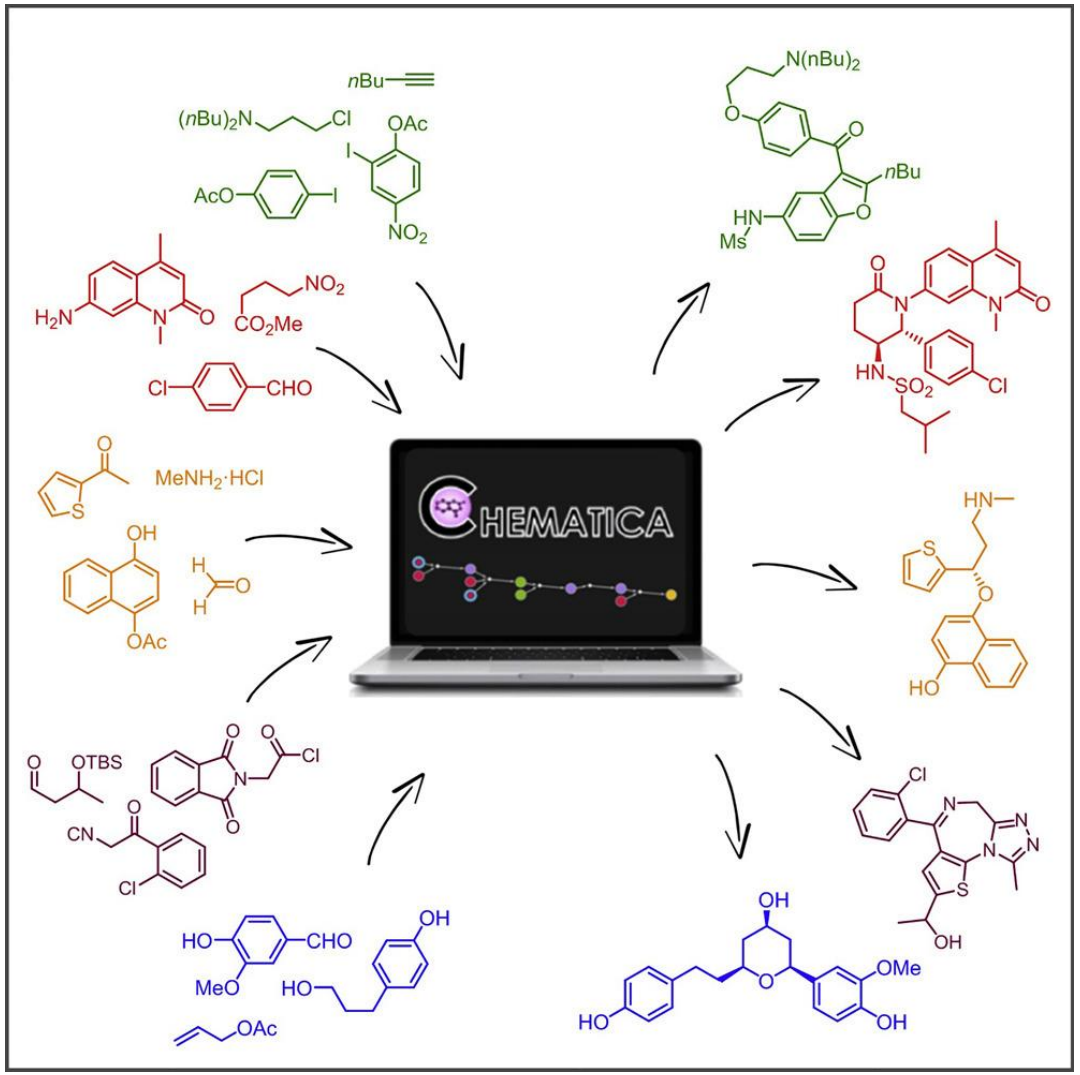
— Chemistry encoded as thousands of hard-coded rules

— Acquired by Merck in 2017, now marketed as Synthia, more frontend development

DL version:

Klucznik T. et al, *Chem*, **2018**, *4*, 522-532. https://doi.org/10.1016/j.chempr.2018.02.002

— Combination of rules and DL networks.



Challenges and discussion points

Open research challenges:

— Incorporating **uncertainty**

— **Physics-informed ML**

— **Autonomous lab**-in-the-loop systems

ML shifts retrosynthesis from

— hard-coded expert rules → data-driven approaches

Discussion points:

— Should ML models suggest reactions they don’t “understand”?

— How do we benchmark creativity vs. safety?

— What is the chemist’s role in all of this?